

KADI SARVA VISHWAVIDHYALAYA

B.E Semester-1 Examination (Branch Name) Month 2023

Subject Code: CC103-N

Subject Name: FUNDAMENTAL OF PROGRAMMING

Date: 30/12/2023

Time: 12:30 PM to 03:30 PM

Total Marks: 70

Instructions:

1. Answer each section in separate answer sheet.
2. Use of scientific calculator is permitted.
3. All questions are Compulsory.
4. Indicate clearly, the option you attempt along with its respective question number.
5. Use the last page of main supplementary for rough work.

Section-I

- Q-1 (A) What is a computer? Draw and explain block diagram of computer in detail. [5]
(B) List out operators used in C language and explain any two with example. [5]
(C) Assume that you want to make sum of 1 to 100. Write a C program to perform the same using (1) While loop (2) Do-while loop. [5]

OR

- (C) Write a C program for simple calculator (+, -, *, /, %) using switch-case statement. [5]
Q-2 (A) Draw and explain structure of a C program. [5]
(B) Write a C program to compute the series $1! + 2! + 3! + \dots + n!$, read value of n from the user. [5]

OR

- Q-2 (A) Write the syntax of else_if ladder and explain with example. [5]
(B) Draw the flowchart write a C program to find the sum of all integers greater than 100 and less than 200 that are divisible by 7. [5]
Q-3 (A) Write a C program to copy string without using strcpy() function. [5]
(B) What is data type? List all types of data type available in C and state the rules to define identifier. [5]

OR

- Q-3 (A) Write a C program to reverse the string without using inbuilt function. [5]
(B) 1. Explain scope and life time of variable. [5]
2. What do you mean by recursive function? Explain with small example.

Section-II

- Q-4 (A) What is string? Explain any two string handling functions with syntax and example. [5]
(B) Write a C program to read 10 numbers from user and store them in an array. Display Sum, Minimum and Average of the numbers. [5]
(C) Explain the function definition, function prototype and function call with relative example. [5]

OR

- (C) Explain one dimensional array, two dimensional array and multi-dimensional array with their syntax and example. [5]

- Q-5 (A) Write a program using structure to get name, roll number, and marks of a student's of a class and find out who got highest marks. [5]
(B) Explain the use of pointer in C programming. How to declare and initialize pointer in C. [5]

OR

- Q-5 (A) Write a C program to display following output using for loop. [5]

```
1
2 1
3 2 1
4 3 2 1
5 4 3 2 1
```

- (B) What is structure? Explain structure within structure with example. [5]
Q-6 (A) Explain the following File Handling functions: [5]

1. fseek()
2. ftell()
3. fread()
4. fwrite()
5. fscanf()

- (B) Explain call by value and call by reference with example. [5]

OR

- Q-6 (A) Explain implicit and explicit type conversion with example. [5]
(B) Write a program to read a matrix of size m*n and print its transpose. [5]